

Fluidodinamica Numerica

Ingegneria Aerospaziale, Università di Napoli ‘Federico II’

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Exercise 27

Streaklines in wake flows

Difficulty: medium-high

Skill required: physical-numerical

Modify the code `TunnelPsiZita.m` by introducing a procedure for the computation of *streaklines* inside the flow.

Streaklines are the curves joining the positions of fluid elements which at some earlier instant passed through the same fixed point in space, as it happens when dye is discharged at some fixed point into moving fluid. From a mathematical point of view, they are defined as the curves joining the points $\mathbf{x}^k(t)$ which are determined by the solution of the systems of ODEs:

$$\frac{d\mathbf{x}^k}{dt} = \mathbf{V}(\mathbf{x}^k(t), t) \quad (1)$$

$$\mathbf{x}^k(t_0 + k\Delta t) = \mathbf{x}_0. \quad (2)$$

The exercise aims at visualize the wake flow past an object by continuously introducing particles at fixed positions upstream the obstacle. The positions of these particles are determined by integrating the system of equations (1)-(2), where the right hand side of Eq. (1) is determined by interpolating at each point $\mathbf{x}^k(t)$ the velocity field $\mathbf{V}$ defined on the grid. The streaklines are then determined by joining the isochronous positions of the particles.